

Demonstration of VirtuEleDent: A Compact XR Tooth-Cutting Training System Using a Physical EMR-based Dental Handpiece and Teeth Model

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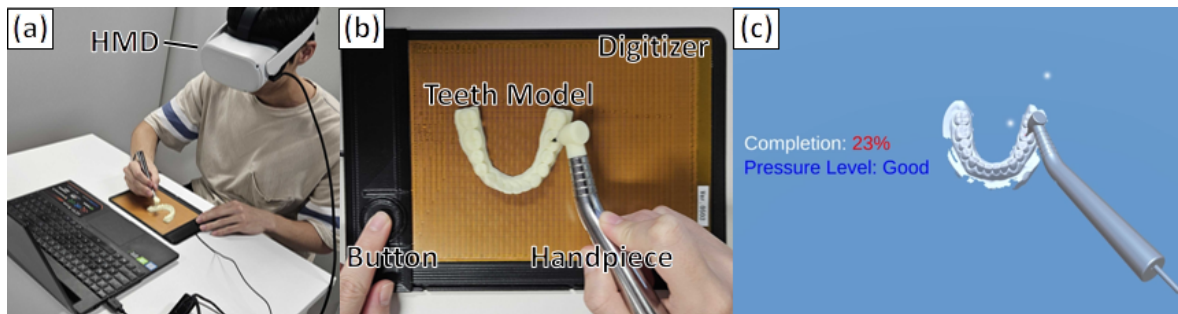


Figure 1: (a) Setup of VirtuEleDent with a physical 3D-printed teeth model and handpiece operated on the EMR-based tracking platform and a mobile virtual HMD. (b) User's handpiece manipulation on the teeth model in the real world. (c) User's view in the virtual environment.

ABSTRACT

Dental cutting is a crucial skill for dental students. However, current VR dental cutting training systems rely on bulky and costly haptic devices, which reduce opportunities for individual practice. Moreover, the limitations imposed by the maximum reaction force of an active haptic device would impact the range of tooth hardness that can be reproduced. We propose a compact XR tooth-cutting training system, VirtuEleDent, that employs a passive haptic approach using a 3D-printed physical teeth model and a three-dimensionally tracked handpiece. Their spatial relationship is accurately rendered in the virtual environment of a mobile head-mounted display (HMD), providing users with realistic haptic sensations during virtual tooth-cutting exercises. Our tracking platform is operated using electromagnetic resonance (EMR) stylus technology and consists of a digitizer (i.e., tracking board) and a handpiece device. A customized EMR stylus unit (i.e., resonance coil) and an inertial measurement unit (IMU) sensor are installed inside the handpiece, allowing for precise measurement of its tip's 3D position and orientation. This setup enables the learner to physically manipulate dexterous handpieces on the teeth model while experiencing virtual tooth-cutting in the HMD.

This is a companion demo to the IEEE VR 2025 Conference paper: "VirtuEleDent: A Compact XR Tooth-Cutting Training System Using a Physical EMR-based Dental Handpiece and Teeth Model." To watch a video about VirtuEleDent, please visit <https://youtu.be/ZvHZ6IEAhYm>.

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[//youtu.be/ZvHZ6IEAhYm](https://youtu.be/ZvHZ6IEAhYm).

Index Terms: Extended reality application, electromagnetic resonance motion tracker

1 INTRODUCTION

Teeth play vital roles in the human body, creating a strong demand for precise dental training. Tooth-cutting is a critical dental skill requiring careful practice to avoid nerve damage and preserve tooth shape. However, traditional training tools, such as SIMROID [2], while highly realistic, are costly, bulky, and require frequent replacement of plastic teeth models.

XR (Extended Reality) training systems offer a promising alternative by enabling virtual or augmented reality-based dental training [1, 4]. These systems allow repetitive training without consuming physical models and provide educational feedback. However, existing VR systems rely on expensive active haptic devices, which have also imposed limitations on accurately simulating the hardness of teeth and their reaction forces, compromising the quality of the training experience.

We propose *VirtuEleDent*, a compact and portable XR tooth-cutting training system using a passive haptic approach. Our system integrates a 3D-printed teeth model and a precisely tracked handpiece device. The handpiece manipulations on the real teeth model are accurately rendered in VR via a head-mounted display (HMD). Focusing on thin-tooth-cutting scenarios, we developed a compact 3D tracking platform based on electromagnetic resonance (EMR) stylus technology. The handpiece device includes a custom EMR stylus unit, an IMU sensor, and a vibrator for enhanced feedback.

2 VIRTUELE DENT

2.1 Design Considerations

To set our goals, we performed iterative interview sessions with two dentistry educators. The timing for interviews was mainly when the COVID-19 infection impacted the educational environment of the university (e.g., no access to the equipment), dramatically highlighting the necessity of portability for individual training systems.

From the interview results, we eventually identified five implementation requirements to meet the increasing demand for compact VR dental training systems.

- The entire setup should be compact and portable while using affordable peripherals.
- The haptic sensation should be realistic in representing the hardness and shape of the teeth.
- The handpiece should be physical and identical in shape to the actual dental handpiece, whose 3D movements are precisely tracked both on and around the teeth model.
- The most common types of tooth-cutting tasks (i.e., on occlusal and lateral (lingual and buccal, labial) surfaces) should be supported.
- The user interface should include basic functions to facilitate repetitive cutting training, including a magnification view and a real-time feedback system.

To embody the design considerations, we attempt a new XR approach, a mix of physical and virtual elements through passive haptic feedback using physical props and a precise 3D tracking system.

2.2 Components and Workflow

VirtuEleDent consists of three main components. The first component, which is a VR HMD that provides learners with an immersive virtual dental training environment rendered on Unity. The second component is a computer software package that manages the sensor data from the tracking platform and the VR training application. The third represents the tangible instruments, which include three parts: (1) a digitizer, the main tracking board of our EMR-based tracking platform, (2) a 3D-printed teeth model (created with the real person's teeth scan data) affixed to the digitizer, and (3) a handpiece device that embeds an EMR tracker at its tip and an IMU sensor inside the body.

Once the digitizer captures the signals from the tip of the handpiece device, its firmware calculates the spatial status of the handpiece (i.e., x, y, z, quaternion, and pressure data) and transmits that data through a USB connection. Similarly, the handpiece's IMU sensor (yaw rotation tracking) data are also sent to the computer through a serial communication. Our sensor data management software first decodes those data from the digitizer and handpiece, which are then sent to the Unity VR training application through a TCP connection. The HMD (Meta Quest 2 or 3) and our Unity software are linked by a Meta Quest Link, supporting both wired and wireless connections.

2.3 VR Training Application

Our training application has a set of fundamental functionalities to facilitate the training experiences. We focused on making essential functions such as cutting and initialization; other minor supplemental updates can be implemented later.

Tooth-cutting tasks are divided into two types: vertical cutting of the occlusal surface and horizontal cutting of the lateral surface using the lateral part of the handpiece bur. The handpiece has a pressure sensor on the tip of the bur to detect the physical contact between the bur tip and the tooth support for occlusal surface cutting. Consequently, both position tracking and pressure sensor data are used in combination to accurately detect bur contact. On the other hand, lateral cutting is performed to form the base of the crown, called the abutment tooth; however, since there is no pressure-sensitive sensor on the side of the handpiece tip, contact with the tooth cannot be detected sensitively by pressure alone. Therefore, in our system, lateral cutting is based on the positional information of the bur. This information is used to calculate the distance between the bur and the tooth in order to determine whether the bur is in contact with the tooth. Unity's Raycast function is then used to pinpoint the cutting area.

2.3.1 Initialization and Functions

We prepared functions specifically for managing our user study and to make them usable for general cavity training scenarios. The cavity treatment function colors part of the occlusal or lateral surfaces in red, and the user can attempt to cut by touching the bur tip or side edge to the colored part. The user's cutting motions are recorded and visualized in blue in real time, and the contact area is modified using vertex-based mesh deformation driven by a compute shader, based on contact time, which is applied to both occlusal and lateral surface cuttings. We also prepare magnification functions that allow for a closer look at the modified part (controlled by the Meta Quest 2 left controller stick). In addition, a reset function is provided to restore the modified (cut) part to the original tooth shape, allowing for repeated cutting training. The target tooth shape can be saved as mesh data if the trainer wants to check the cutting performance after all of the training is finished.

In place of the foot pedal used in a dental clinic, a button on the digitizer can be used to control the on/off switching of the cutting operation. While this button is pressed, the handpiece vibrates, a motor sound is emitted, and cutting can proceed. A cutting sound is emitted in addition to the motor sound when the bur tip contacts the tooth and cutting is carried out.

The pressure information presentation system indicates to the user whether the force applied is appropriate through a text display and a visual effect that flashes a red light in the user's field of vision. This alert is activated based on the pressure sensor of the bur tip during the occlusal surface cutting process. This allows the user to learn the amount of appropriate force for cutting. The threshold can be customized differently to simulate various tooth hardnesses. Since the appropriate pressure depends on the hardness of the teeth (depending on the patient), our current parameter (150 gf) was set based on empirical recommendations from dentists based on the hardness of the teeth model used in this prototype.

3 CONCLUSION

In this demonstration, we show a set of fundamental tooth-cutting scenarios where a highly accurate tracking system and realistic haptic feedback can be experienced in our compact and affordable setup, which will initiate active discussions with visitors in both practical and technical aspects.

We direct readers to our paper [3] for further analysis, including expert user studies and discussion around the system's potential, acceptance, and improvements from an educator's perspective.

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